

Schatten rigidity for scaled Fock quantization

A sharp-range answer to the final problem in arXiv:2205.12345

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Status. High-confidence candidate partial result, pending human review. It completely answers the Hilbert–Schmidt question in complex dimension $n \geq 2$, the trace-class question in every dimension, and more generally every Schatten exponent $1 \leq p \leq n$. The range $p > n$ remains open by this method.

The source question

Let $\varphi \in C^2(\mathbb{C}^n)$ be real-valued with $i\partial\bar{\partial}\varphi \simeq \omega_0$, and put

$$d\mu_t(z) = t^{-n} e^{-2\varphi(z/\sqrt{t})} dv(z), \quad \mathcal{F}_t^2(\varphi) = H(\mathbb{C}^n) \cap L^2(\mu_t).$$

Writing $T_u^{(t)} = P^{(t)} M_u$, the source asks for the bounded f such that, for every bounded g ,

$$\|T_f^{(t)} T_g^{(t)} - T_{fg}^{(t)}\|_{S_p} \longrightarrow 0 \quad (t \downarrow 0), \tag{1}$$

especially for the Hilbert–Schmidt norm $p = 2$.

if and only if $g \in \text{VMO}$. Here $\|\cdot\| = \|\cdot\|_{F_t^2(\varphi) \rightarrow F_t^2(\varphi)}$. For further details, see [\[13\]](#).

As for an open problem in this line of work, it would be interesting to characterize those symbols $f \in L^\infty(\mathbb{C}^n)$ for which [\(13\)](#) holds for all $g \in L^\infty(\mathbb{C}^n)$ when the operator norm is replaced by the Hilbert–Schmidt (or other Schatten class) norm.

ACKNOWLEDGMENTS

The exact page is bundled as `source_excerpt_open_problem_page_10.pdf`.

Theorem 1 (sharp-range characterization). *Let $1 \leq p \leq n$ and $f \in L^\infty(\mathbb{C}^n)$. Then (1) holds for every $g \in L^\infty(\mathbb{C}^n)$ if and only if f is constant almost everywhere. Consequently:*

- the Hilbert–Schmidt answer is exactly the constants when $n \geq 2$;
- the trace-class answer is exactly the constants for every $n \geq 1$.

Schatten/IDA input and its scaling

For $u \in L_{\text{loc}}^2(\mathbb{C}^n)$ define

$$G_r(u)(z) = \inf_{h \in H(B(z,r))} \left(\frac{1}{|B(z,r)|} \int_{B(z,r)} |u - h|^2 dv \right)^{1/2}.$$

Hu–Virtanen’s Theorem 1.1 states, for every $q > 0$,

$$\|H_u\|_{S_q} \asymp \|G_1(u)\|_{L^q(\mathbb{C}^n)}. \quad (\text{HV})$$

It applies here because L^∞ lies in their symbol class $\mathcal{S}$. The constants depend on q, n, φ , but not on u .

For $r = \sqrt{t}$, the unitary dilation $U_t h(z) = h(rz)$ maps the scaled Fock space to the fixed space and conjugates $H_u^{(t)}$ to H_{u_r} , where $u_r(z) = u(rz)$. Moreover

$$G_1(u_r)(z) = G_r(u)(rz).$$

Since $\mathbb{C}^n$ has real dimension $2n$, (HV) therefore gives the exact scaling law, up to fixed comparison constants,

$$\|H_u^{(t)}\|_{S_{2p}} \asymp r^{-n/p} \|G_r(u)\|_{L^{2p}(\mathbb{C}^n)}. \quad (1)$$

A rigidity lemma

Lemma 1. *Let $q \geq 1$ and $u \in L^2_{\text{loc}}(\mathbb{C}^n)$. If*

$$\|G_r(u)\|_{L^q} = o(r) \quad (r \downarrow 0),$$

then u agrees almost everywhere with an entire holomorphic function.

Proof. Choose $\eta \in C_c^\infty(B(0, 1))$ with integral one and put $\eta_r(x) = r^{-2n}\eta(x/r)$. At each x , choose a holomorphic near-minimizer h_x in the definition of $G_r(u)(x)$. Integration by parts is legitimate because $\eta_r(x - \cdot)$ is compactly supported in $B(x, r)$, and $\bar{\partial}h_x = 0$. Thus Cauchy–Schwarz gives, component by component,

$$\begin{aligned} |\bar{\partial}(u * \eta_r)(x)| &= \left| \int_{B(x, r)} (u(y) - h_x(y)) \bar{\partial}_x \eta_r(x - y) dv(y) \right| \\ &\leq Cr^{-1} G_r(u)(x). \end{aligned}$$

Indeed, the local L^2 error contributes $r^n G_r(u)(x)$, while $\|\nabla \eta_r\|_2 = O(r^{-n-1})$. Hence

$$\|\bar{\partial}(u * \eta_r)\|_{L^q} \leq Cr^{-1} \|G_r(u)\|_{L^q} \longrightarrow 0.$$

The mollifications converge to u locally in distributions, so $\bar{\partial}u = 0$ distributionally. Weyl’s lemma for $\bar{\partial}$ finishes the proof. $\square$

Proof of Theorem 1

The Fock product identity is

$$T_f^{(t)} T_g^{(t)} - T_{fg}^{(t)} = -(H_{\bar{f}}^{(t)})^* H_g^{(t)}.$$

Use the universal quantifier in (1) only once, with $g = \bar{f}$. For every bounded operator H , membership of $H^* H$ in S_p is equivalent to membership of H in S_{2p} , and

$$\|H^* H\|_{S_p} = \|H\|_{S_{2p}}^2.$$

Consequently (1) forces $\|H_{\bar{f}}^{(t)}\|_{S_{2p}} \rightarrow 0$. Applying (1) with $u = \bar{f}$ yields

$$\|G_r(\bar{f})\|_{L^{2p}} = o(r^{n/p}). \quad (2)$$

If $p \leq n$, then $n/p \geq 1$, and (2) is in particular $o(r)$. Lemma 1 says that $\bar{f}$ is entire. It is bounded, hence constant by Liouville’s theorem, and so is f .

Conversely, if $f = c$ almost everywhere, then $T_f^{(t)} = cI$ and $T_f^{(t)} T_g^{(t)} - T_{fg}^{(t)} = 0$ for every g and t . $\square$

Proof intuition, sharpness, and remaining range

The single adversarial choice $g = \bar{f}$ converts the semicommutator into a positive Hankel square. Schatten/IDA theory then translates small quantum defect into the analytic-approximation rate $o(r^{n/p})$. The geometric threshold is exactly where that rate reaches first order: $p \leq n$. First-order decay kills $\bar{\partial}\bar{f}$ through mollification, leaving a bounded entire function and therefore a constant.

For $p > n$, the forced exponent n/p is below one, so the derivative estimate need not vanish. This is a genuine obstruction to this route: a nonconstant smooth compactly supported u has $G_r(u) = O(r)$ locally and hence passes the necessary $g = \bar{f}$ test in that range. The quantifier over all rough g may still impose more, but no complete proof or counterexample was found. Thus the packet does not claim the Hilbert–Schmidt answer in dimension one.

The result also separates the Schatten problem from the operator-norm answer: nonconstant compactly supported smooth symbols lie in VMO, yet Theorem 1 excludes them for Hilbert–Schmidt convergence when $n \geq 2$.

References

- [1] J. A. Virtanen, *On the product formula for Toeplitz and related operators*, arXiv:2205.12345, Section 5 and p. 10.
- [2] Z. Hu and J. A. Virtanen, *Schatten class Hankel operators on the Segal–Bargmann space and the Berger–Coburn phenomenon*, arXiv:2012.13768v2, Theorem 1.1.
- [3] Z. Hu and J. A. Virtanen, *Corrigendum to “Schatten class Hankel operators on the Segal–Bargmann space and the Berger–Coburn phenomenon”*, arXiv:2012.13768v4. The corrigendum changes Theorems 1.2 and 2.6, not the Theorem 1.1 used above.